INCH-POUND

MIL-DTL-5757/13J
13 July 2022
SUPERSEDING
MIL-R-5757/13H
3 September 2004

DETAIL SPECIFICATION SHEET

# RELAYS, ELECTRICAL, HERMETICALLY SEALED, DPDT, LOW LEVEL TO 2 AMPERES, (SENSITIVE, 40 MILLIWATTS)

INACTIVE FOR NEW DESIGN
AFTER 15 OCTOBER 1998

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-5757.

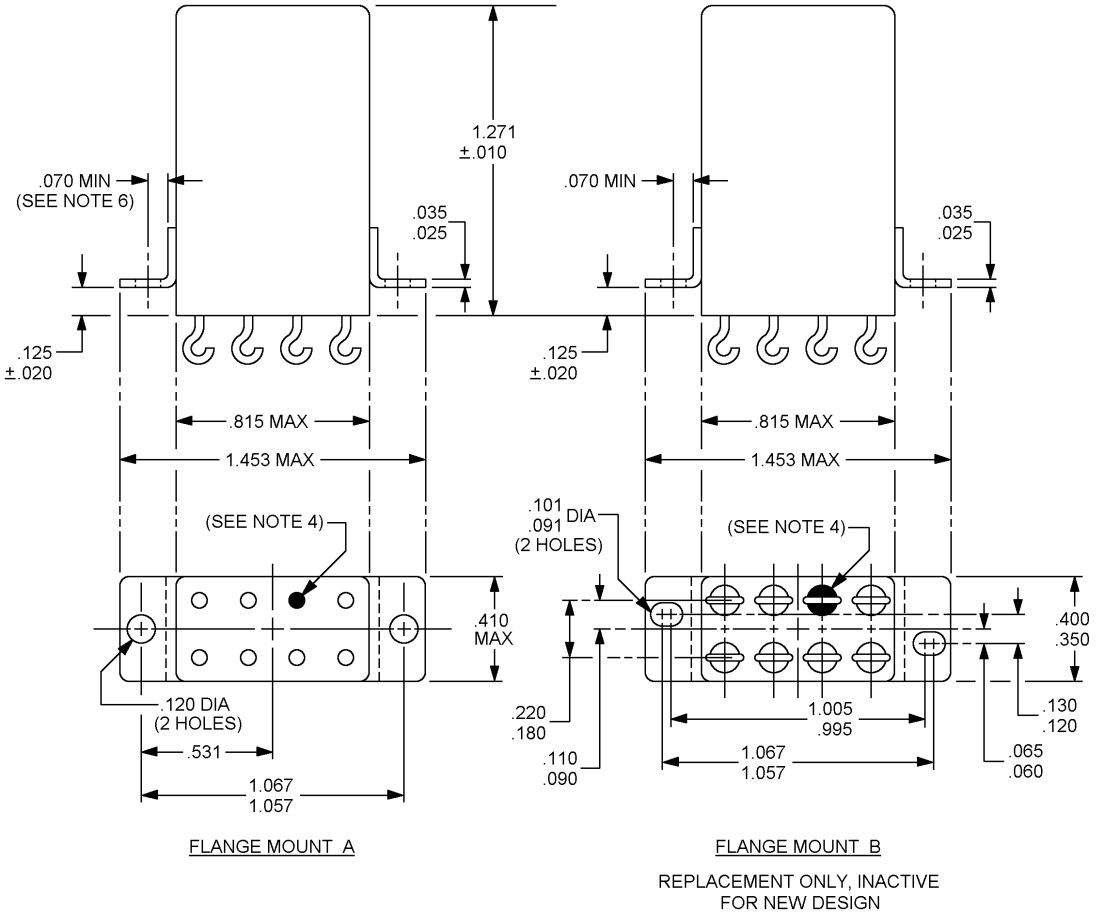



FIGURE 1. Dimensions and configuration.

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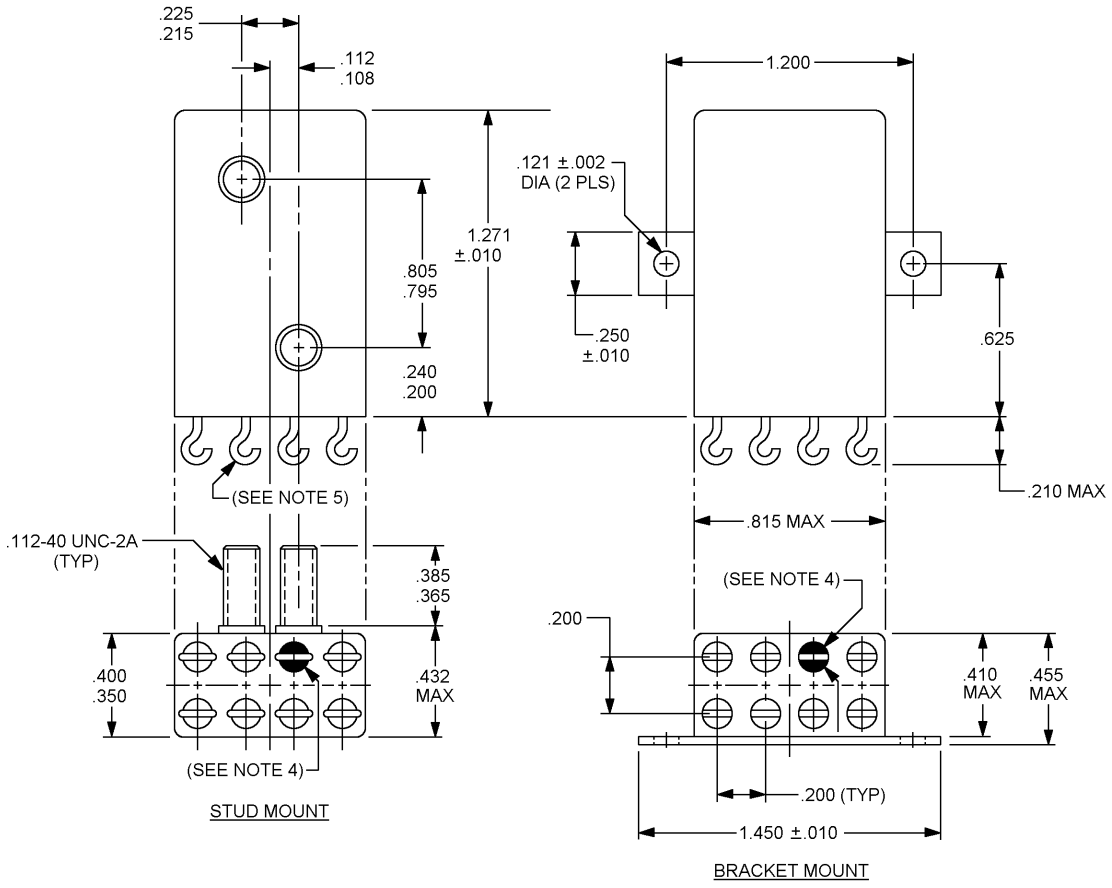
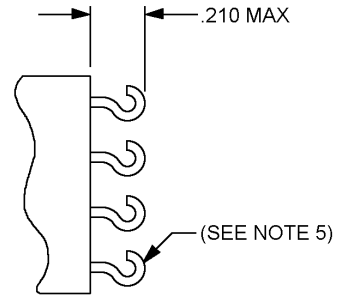


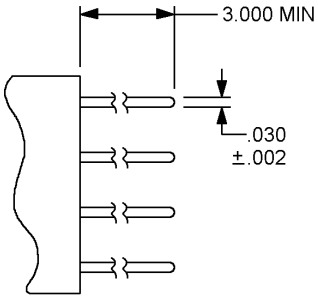

FIGURE 1. Dimensions and configuration - Continued.



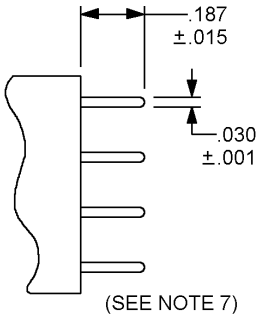
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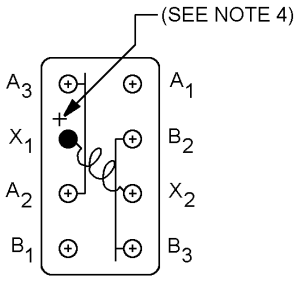

SOLDER LUG



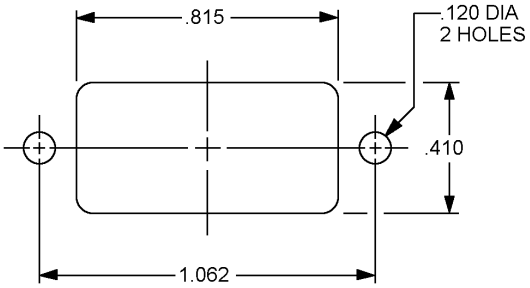

WIRE LEAD



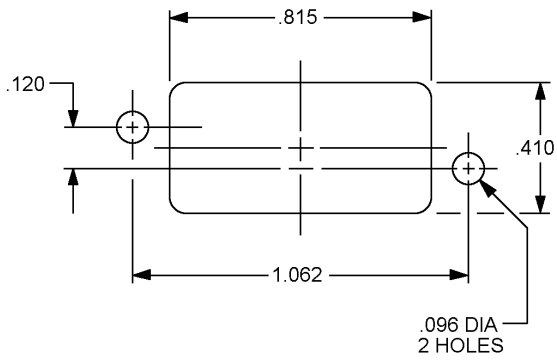

PIN




SCHEMATIC DIAGRAM UNENERGIZED POSITION
BOTTOM VIEW




FLANGE MOUNT A
CHASSIS CUTOUT




RECOMMENDED FLANGE MOUNT B
CHASSIS CUTOUT

FIGURE 1. Dimensions and configuration - Continued.



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .001 | 0.03 | .101 | 2.57 | .225 | 5.72 | .815 | 20.70 |
| .002 | 0.05 | .108 | 2.74 | .240 | 6.10 | .995 | 25.27 |
| .010 | 0.25 | .110 | 2.79 | .250 | 6.35 | 1.005 | 25.53 |
| .015 | 0.38 | .112 | 2.84 | .350 | 8.89 | 1.057 | 26.85 |
| .020 | 0.51 | .120 | 3.05 | .365 | 9.27 | 1.062 | 26.97 |
| .025 | 0.64 | .121 | 3.07 | .385 | 9.78 | 1.067 | 27.10 |
| .030 | 0.76 | .125 | 3.18 | .400 | 10.16 | 1.200 | 30.48 |
| .035 | 0.89 | .130 | 3.30 | .410 | 10.41 | 1.271 | 32.28 |
| .060 | 1.52 | .180 | 4.57 | .432 | 10.97 | 1.450 | 36.83 |
| .065 | 1.65 | .187 | 4.75 | .455 | 11.56 | 1.453 | 36.91 |
| .070 | 1.78 | .200 | 5.08 | .531 | 13.49 | 3.000 | 76.20 |
| .090 | 2.29 | .210 | 5.33 | .625 | 15.88 | | |
| .091 | 2.31 | .215 | 5.46 | .795 | 20.19 | | |
| .096 | 2.36 | .220 | 5.59 | .805 | 20.45 | | |

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
3. Metric equivalents are given for general information only.
4. The $X_1$ terminal shall be identified with a contrasting bead. In the case of relays with polarized coils, the contrasting bead shall indicate the positive terminal and a +(plus) sign shall be placed on the schematic diagram to indicate this terminal.
5. Shape of solder terminals optional.
6. Mounting screw head clearance for .995 (25.27 mm) -1.005 (25.53 mm) dimension based on use of No. 2 filister head screw.
7. Relays with pin terminals shall not be used for plug-in socket applications.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Contact data:

Configuration: DPDT

Arrangement: 2 form C

Load ratings: 1/

High level (relay case grounded):

Resistive: 2 amperes at 28 V dc.

0.3 ampere at 115 V ac, 60 to 400 Hz. 2/

Inductive: 0.75 ampere at 28 V dc, at an inductive load of 200 millihenries, minimum.

Lamp: 0.1 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Rated life:

Before: 0.05 ohm.

During: 10 percent of open circuit voltage maximum.

After: 0.1 ohm.

Intermediate current:

Before: 0.05 ohm.

During: 3 ohms.

After: 3 ohms.

Low level:

Before: 0.05 ohm.

During: 100 ohms.

After: 0.15 ohm.

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1/ Each relay possesses high and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum, or peak ac open circuit, are not recommended for subsequent use in low level applications.

2/ For qualifications, testing of 400 hertz (Hz) loads are not required.



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Contact bounce: 2 milliseconds (ms) maximum, load shall be 10 mA at 6 V dc maximum.

Overload test:

Resistive: 2 times rated current.

Inductive: 2 times rated current.

COIL DATA:

Duty rating: Continuous.

Nominal current: See table I.

Pick-up current: See table I.

Drop-out current: See table I.

Coil release: See table I.

Operate time: 15 ms maximum over temperature range.

Release time: 10 ms maximum over temperature range.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms minimum except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

| | Sea level V rms (60 Hz) | Altitude V rms (60 Hz) |
|---|---|---|
| Between case, frame or enclosure, and all contacts both in the energized and deenergized positions: ---------- | 1,000 | 350 All terminals to case |
| Between case, frame or enclosure and coil: ---------- | 500 | |
| Between all contacts and coil: ---------- | 1,000 | |
| Between open contacts in the energized and deenergized positions: ---------- | 500 | |
| Between contact poles: ---------- | 1,000 | |

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration: 15 g's.

Acceleration: 30 g's.

Shock: 100 g's.

Resistance to soldering heat: Applicable to pin type (printed circuit).

Internal moisture: Applicable.



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PHYSICAL:

Enclosure design: Hermetically sealed.

Terminal strength: 3 ±0.3 pound pull.

Terminal twist test: Applicable to wire leads.

Sealed by welding: Applicable.

Dimensions and configuration: See figure 1.

Termination: See figure 1 and table I.

Weight: 1.1 ounces maximum.

Part or Identifying Number (PIN): See table I.

LIFE TEST REQUIREMENTS:

High level: 100,000 operations each at rated load.

Two relays per contact rating shall be tested, with rated loads on all contacts, except intermediate current.

For qualification inspection, life lamp load test not required.

Low level: Applicable (100,000 cycles).

Intermediate current: (50,000 cycles).

VERIFICATION (See table II):

Group A.

a. Group A2: 100 percent.

b. Dielectric withstanding voltage:

(1) Tests to be conducted at sea level rating only.

(2) Duration of application: 5-10 seconds at a 10 percent increase in the dielectric strength voltage.



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TABLE I. PIN and characteristics.

| PIN M5757/13- | Mount | Terminal | DC coil resistance (±10%) +25°C ±5°C (ohms) | Nominal coil current (mA) | Pickup current (max) (mA) | Dropout current | |
|---|---|---|---|---|---|---|---|
| | | | | | | Max (mA) | Min (mA) |
| 083 | Flange A | Lug (solder) | 20.0 | 90.0 | 45.0 | 22.5 | 4.5 |
| 084 | Flange A | Lug (solder) | 100.0 | 40.0 | 20.0 | 10.0 | 2.0 |
| 085 | Flange A | Lug (solder) | 500.0 | 18.0 | 9.0 | 4.5 | 0.9 |
| 086 | Flange A | Lug (solder) | 1,000 | 13.0 | 6.5 | 3.1 | 0.65 |
| 087 | Flange A | Lug (solder) | 1,500 | 11.0 | 5.2 | 2.6 | 0.52 |
| 088 | Flange A | Lug (solder) | 2,000 | 9.0 | 4.5 | 2.2 | 0.50 |
| 089 | Flange A | Lug (solder) | 2,500 | 8.0 | 4.0 | 2.0 | 0.40 |
| 090 | Flange A | Lug (solder) | 5,000 | 6.0 | 2.8 | 1.4 | 0.30 |
| 091 | Flange A | Lug (solder) | 8,000 | 5.0 | 2.3 | 1.1 | 0.23 |
| 092 | Flange A | Lug (solder) | 10,000 | 4.0 | 2.0 | 1.0 | 0.20 |
| 093 | Flange A | Pin (PW) | 20.0 | 90.0 | 45.0 | 22.5 | 4.5 |
| 094 | Flange A | Pin (PW) | 100.0 | 40.0 | 20.0 | 10.0 | 2.0 |
| 095 | Flange A | Pin (PW) | 200.0 | 28.4 | 14.2 | 10.0 | 1.4 |
| 096 | Flange A | Pin (PW) | 500.0 | 18.0 | 9.0 | 4.5 | 0.9 |
| 097 | Flange A | Pin (PW) | 1,000 | 13.0 | 6.5 | 3.1 | 0.65 |
| 098 | Flange A | Pin (PW) | 1,500 | 11.0 | 5.2 | 2.6 | 0.52 |
| 099 | Flange A | Pin (PW) | 2,000 | 9.0 | 4.5 | 2.2 | 0.50 |
| 100 | Flange A | Pin (PW) | 2,500 | 8.0 | 4.0 | 2.0 | 0.40 |
| 101 | Flange A | Pin (PW) | 5,000 | 6.0 | 2.8 | 1.4 | 0.30 |
| 102 | Flange A | Pin (PW) | 8,000 | 5.0 | 2.3 | 1.1 | 0.23 |
| 103 | Flange A | Pin (PW) | 10,000 | 4.0 | 2.0 | 1.0 | 0.20 |
| 104 | Flange B | Lug (solder) | 20.0 | 90.0 | 45.0 | 22.5 | 4.5 |
| 105 | Flange B | Lug (solder) | 100.0 | 40.0 | 20.0 | 10.0 | 2.0 |
| 106 | Flange B | Lug (solder) | 500.0 | 18.0 | 9.0 | 4.5 | 0.9 |
| 107 | Flange B | Lug (solder) | 1,000 | 13.0 | 6.5 | 3.1 | 0.65 |
| 108 | Flange B | Lug (solder) | 1,500 | 11.0 | 5.2 | 2.6 | 0.52 |
| 109 | Flange B | Lug (solder) | 2,000 | 9.0 | 4.5 | 2.2 | 0.50 |
| 110 | Flange B | Lug (solder) | 2,500 | 8.0 | 4.0 | 2.0 | 0.40 |
| 111 | Flange B | Lug (solder) | 5,000 | 6.0 | 2.8 | 1.4 | 0.30 |
| 112 | Flange B | Lug (solder) | 8,000 | 5.0 | 2.3 | 1.1 | 0.23 |
| 113 | Flange B | Lug (solder) | 10,000 | 4.0 | 2.0 | 1.0 | 0.20 |
| 114 | Stud | Lug (solder) | 20.0 | 90.0 | 45.0 | 22.5 | 4.5 |
| 115 | Stud | Lug (solder) | 100.0 | 40.0 | 20.0 | 10.0 | 2.0 |
| 116 | Stud | Lug (solder) | 500.0 | 18.0 | 9.0 | 4.5 | 0.9 |
| 117 | Stud | Lug (solder) | 1,000 | 13.0 | 6.5 | 3.1 | 0.65 |



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TABLE I. PIN and characteristics - Continued.

| PIN M5757/13- | Mount | Terminal | DC coil resistance (±10%) +25°C ±5°C (ohms) | Nominal coil current (mA) | Pickup current (max) (mA) | Dropout current | |
|---|---|---|---|---|---|---|---|
| | | | | | | Max (mA) | Min (mA) |
| 118 | Stud | Lug (solder) | 1,500 | 11.0 | 5.2 | 2.6 | 0.52 |
| 119 | Stud | Lug (solder) | 2,000 | 9.0 | 4.5 | 2.2 | 0.50 |
| 120 | Stud | Lug (solder) | 2,500 | 8.0 | 4.0 | 2.0 | 0.40 |
| 121 | Stud | Lug (solder) | 5,000 | 6.0 | 2.8 | 1.4 | 0.30 |
| 122 | Stud | Lug (solder) | 8,000 | 5.0 | 2.3 | 1.1 | 0.23 |
| 123 | Stud | Lug (solder) | 10,000 | 4.0 | 2.0 | 1.0 | 0.20 |
| 124 | Stud | Wire lead | 20.0 | 90.0 | 45.0 | 22.5 | 4.5 |
| 125 | Stud | Wire lead | 100.0 | 40.0 | 20.0 | 10.0 | 2.0 |
| 126 | Stud | Wire lead | 500.0 | 18.0 | 9.0 | 4.5 | 0.9 |
| 127 | Stud | Wire lead | 1,000 | 13.0 | 6.5 | 3.1 | 0.65 |
| 128 | Stud | Wire lead | 1,500 | 11.0 | 5.2 | 2.6 | 0.52 |
| 129 | Stud | Wire lead | 2,000 | 9.0 | 4.5 | 2.2 | 0.50 |
| 130 | Stud | Wire lead | 2,500 | 8.0 | 4.0 | 2.0 | 0.40 |
| 131 | Stud | Wire lead | 5,000 | 6.0 | 2.8 | 1.4 | 0.30 |
| 132 | Stud | Wire lead | 8,000 | 5.0 | 2.3 | 1.1 | 0.23 |
| 133 | Stud | Wire lead | 10,000 | 4.0 | 2.0 | 1.0 | 0.20 |
| 134 | Bracket | Lug solder | 20.0 | 90.0 | 45.0 | 22.5 | 4.5 |
| 135 | Bracket | Lug solder | 100.0 | 40.0 | 20.0 | 10.0 | 2.0 |
| 136 | Bracket | Lug solder | 500.0 | 18.0 | 9.0 | 4.5 | 0.9 |
| 137 | Bracket | Lug solder | 1,000 | 13.0 | 6.5 | 3.1 | 0.65 |
| 138 | Bracket | Lug solder | 1,500 | 11.0 | 5.2 | 2.6 | 0.52 |
| 139 | Bracket | Lug solder | 2,000 | 9.0 | 4.5 | 2.2 | 0.50 |
| 140 | Bracket | Lug solder | 2,500 | 8.0 | 4.0 | 2.0 | 0.40 |
| 141 | Bracket | Lug solder | 5,000 | 6.0 | 2.8 | 1.4 | 0.30 |
| 142 | Bracket | Lug solder | 8,000 | 5.0 | 2.3 | 1.1 | 0.23 |
| 143 | Bracket | Lug solder | 10,000 | 4.0 | 2.0 | 1.0 | 0.20 |



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TABLE II. Qualification inspection and sample size.

| Single submission | Gro up submission | |
|---|---|---|
| 20 units plus 1 open unit.<br>Table I<br>Qualification inspection as applicable. | M5757/13-092 | 20 units plus 1 open unit.<br>Qualification inspection as applicable. |
| | M5757/13-113<br>M5757/13-133<br>M5757/13-143 | 2 units each PIN<br>Qualification inspection Q1 plus shock, vibration, acceleration, terminal strength, and seal. |
| | M5757/13-103 | 4 units<br>Qualification inspection and shock, vibration, acceleration, terminal strength, resistance to soldering heat, and seal. |



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SUPERSESSION DATA:

Supersession data: See table III.

TABLE III. Supersession data.

| Superseded PIN M5757/13- | New PIN M5757/13- | Superseded PIN M5757/13- | New PIN M5757/13- |
|---|---|---|---|
| 011 | 104 | 054 | 097 |
| 012 | 105 | 055 | 098 |
| 013 | 106 | 056 | 099 |
| 014 | 107 | 057 | 100 |
| 015 | 108 | 058 | 101 |
| 016 | 109 | 059 | 102 |
| 017 | 110 | 060 | 103 |
| 018 | 111 | 061 | 124 |
| 019 | 112 | 062 | 125 |
| 020 | 113 | 063 | 126 |
| 031 | 114 | 064 | 127 |
| 032 | 115 | 065 | 128 |
| 033 | 116 | 066 | 129 |
| 034 | 117 | 067 | 130 |
| 035 | 118 | 068 | 131 |
| 036 | 119 | 069 | 132 |
| 037 | 120 | 070 | 133 |
| 038 | 121 | 072 | 134 |
| 039 | 122 | 073 | 135 |
| 040 | 123 | 074 | 136 |
| 041 | 083 | 075 | 137 |
| 042 | 084 | 076 | 138 |
| 043 | 085 | 077 | 139 |
| 044 | 086 | 078 | 140 |
| 045 | 087 | 079 | 141 |
| 046 | 088 | 080 | 142 |
| 047 | 089 | 081 | 143 |
| 048 | 090 | 082 | 095 |
| 049 | 091 | | |
| 050 | 092 | | |
| 051 | 093 | | |
| 052 | 094 | | |
| 053 | 096 | | |

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